

MongoDB Project

1- In Visual studio code, in mongodb_project, after opening mongo.js file and run in vs terminal:

node mongo.js

the result in the terminal:

```
PS D:\Sec. Mis Project> cd mongodb_project
PS D:\Sec. Mis Project\mongodb_project> node mongo.js
Connected to MongoDB!
Students inserted
Courses inserted
Deleted one student
Deleted one course
Score array added to students
Score array added to courses
Score positions updated for students
Score positions updated for courses
Score multiplied by 20 for students
Score multiplied by 20 for courses
Enrollments (One-to-Many) inserted

Final Students:
[
  {
    _id: 1,
    name: 'Ahmed',
    age: 20,
    grade: 'A',
    Score: [ 200, 400, 100, 800, 1000 ]
  },
  {
    _id: 2,
    name: 'Sara',
    age: 22,
    grade: 'B',
    Score: [ 200, 400, 600, 120, 1000 ]
  }
]
```

```
Final Courses:
[
  {
    _id: 1,
    title: 'Databases',
    credit: 3,
    Score: [ 200, 400, 100, 800, 1000 ]
  },
  {
    _id: 2,
    title: 'Networking',
    credit: 3,
    Score: [ 200, 400, 600, 120, 1000 ]
  }
]

Enrollments:
[
  { _id: 1, student_id: 1, course_id: 1 },
  { _id: 2, student_id: 1, course_id: 2 },
  { _id: 3, student_id: 2, course_id: 1 }
]
```

2- In the power shell, run this commands line by line:

```
docker exec -it my_mongo mongosh  
use MIS_DB
```

And then write the Aggregation function:

```
db.enrollments.aggregate([
  {
    $lookup: {
      from: "students",
      localField: "student_id",
      foreignField: "_id",
      as: "student_info"
    }
  },
  {
    $lookup: {
      from: "courses",
      localField: "course_id",
      foreignField: "_id",
      as: "course_info"
    }
  },
  {
    $project: {
      _id: 1,
      student_info: { $arrayElemAt: ["$student_info", 0] },
      course_info: { $arrayElemAt: ["$course_info", 0] }
    }
  }
])
```

The result in the power shell after run the Aggregation function:

```
[
  {
    _id: 1,
    student_info: {
      _id: 1,
      name: 'Ahmed',
      age: 20,
      grade: 'A',
      Score: [ 200, 400, 100, 800, 1000 ]
    },
    course_info: {
      _id: 1,
      title: 'Databases',
      credit: 3,
      Score: [ 200, 400, 100, 800, 1000 ]
    }
  },
  {
    _id: 2,
    student_info: {
      _id: 1,
      name: 'Ahmed',
      age: 20,
      grade: 'A',
      Score: [ 200, 400, 100, 800, 1000 ]
    },
    course_info: {
      _id: 2,
      title: 'Networking',
      credit: 3,
      Score: [ 200, 400, 600, 120, 1000 ]
    }
  },
  {
    _id: 3,
    student_info: {
      _id: 2,
      name: 'Sara',
      age: 22,
      grade: 'B',
      Score: [ 200, 400, 600, 120, 1000 ]
    },
    course_info: {
      _id: 1,
      title: 'Databases',
      credit: 3,
      Score: [ 200, 400, 100, 800, 1000 ]
    }
  }
]
MIS_DB> |
```

Cassandra Project

1-In Visual studio code, in cassandra_project, after opening cassandra.js file and run in vs terminal:

node cassandra.js

the result in the visual studio terminal:

```
PS D:\Sec. Mis Project\cassandra_project> node cassandra.js
Connected to Cassandra!
Keyspace created
Table created
5 rows inserted
Row updated
Row deleted

Final table:
Row {
  department: 'IT',
  student_id: 4,
  age: 23,
  grade: 'C',
  name: 'Layla'
}
Row {
  department: 'CS',
  student_id: 1,
  age: 20,
  grade: 'A',
  name: 'Ahmed'
}
Row {
  department: 'CS',
  student_id: 2,
  age: 22,
  grade: 'A+',
  name: 'Sara'
}
Row {
  department: 'CS',
  student_id: 3,
  age: 21,
  grade: 'A',
  name: 'Omar'
}
```

2-In the power shell, run this commands line by line:

```
docker exec -it my_cassandra cqlsh
```

```
USE mis_keyspace;
```

continue in the CMD and write (all the lines in one time):

```
CREATE TABLE IF NOT EXISTS students (  
    department TEXT,  
    student_id INT,  
    name      TEXT,  
    age       INT,  
    grade     TEXT,  
    PRIMARY KEY ((department), student_id)  
);
```

continue in the CMD and write (all the lines in one time):

```
INSERT INTO students (department, student_id, name, age, grade) VALUES ('CS', 1, 'Ahmed', 20,  
'A');
```

```
INSERT INTO students (department, student_id, name, age, grade) VALUES ('CS', 2, 'Sara', 22, 'B');
```

```
INSERT INTO students (department, student_id, name, age, grade) VALUES ('CS', 3, 'Omar', 21,  
'A');
```

```
INSERT INTO students (department, student_id, name, age, grade) VALUES ('IT', 4, 'Layla', 23, 'C');
```

```
INSERT INTO students (department, student_id, name, age, grade) VALUES ('IT', 5, 'Khaled', 20,  
'B');
```

continue in the CMD and write (all the lines in one time):

```
SELECT * FROM students  
WHERE department = 'CS'  
ORDER BY student_id DESC;
```

And the result in the power shell:

```
cqlsh:mis_keyspace> SELECT * FROM students  
... WHERE department = 'CS'  
... ORDER BY student_id DESC;  
  
department | student_id | age | grade | name  
-----  
CS | 3 | 21 | A | Omar  
CS | 2 | 22 | B | Sara  
CS | 1 | 20 | A | Ahmed  
  
(3 rows)
```

continue in the CMD and write (line by line):

```
CREATE INDEX ON students (grade); (if there is no index)
```

```
SELECT * FROM students WHERE grade = 'A';
```

And the result in the Power Shell:

```
students_grade_idx"  
cqlsh:mis_keyspace> SELECT * FROM students WHERE grade = 'A';  
  
department | student_id | age | grade | name  
-----  
CS | 1 | 20 | A | Ahmed  
CS | 3 | 21 | A | Omar  
  
(2 rows)
```

NEO4J Project

1- Go to Neo4j Desktop and run the instance

2-In Visual studio code, in neo4j_project, after opening cassandra.js file and run in vs terminal:

node neo4j.js

And the result in the visual studio terminal:

```
PS D:\Sec. Mis Project\neo4j_project> node neo4j.js
Connected to Neo4j!
Old data cleared
Graph created with nodes and relationships
Deleted ENEMY_OF relationship between Draco and Harry
Deleted Voldemort node
Removed 'house' property from Draco
Updated Harry's properties
Updated friendship strength relationship

Characters in Gryffindor:
- Harry Potter, Age: 18, House: Gryffindor
- Hermione Granger, Age: 17, House: Gryffindor
- Ron Weasley, Age: 17, House: Gryffindor
- Dumbledore, Age: 115, House: Gryffindor

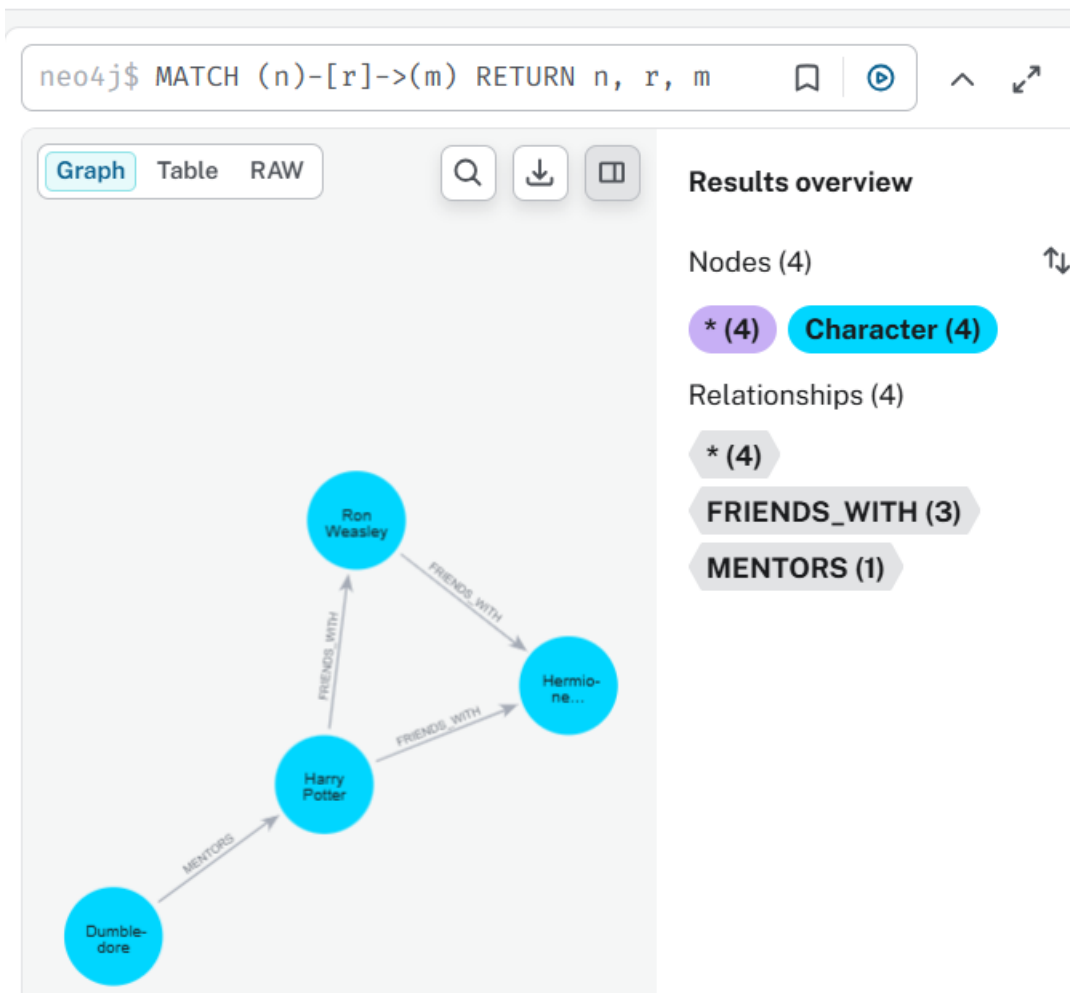
Strong Friendships (strength >= 9):
- Harry Potter → Hermione Granger | Strength: 10 | Since: Year 1
- Harry Potter → Ron Weasley | Strength: 10 | Since: Year 1
❖ PS D:\Sec. Mis Project\neo4j_project>
```


3- go to neo4j desktop, to the instance, click at connect dropdown and choose query

to the query tab write this commands (line by line) to see the graph and the table:

MATCH (n)-[r]->(m) RETURN n, r, m

The result:



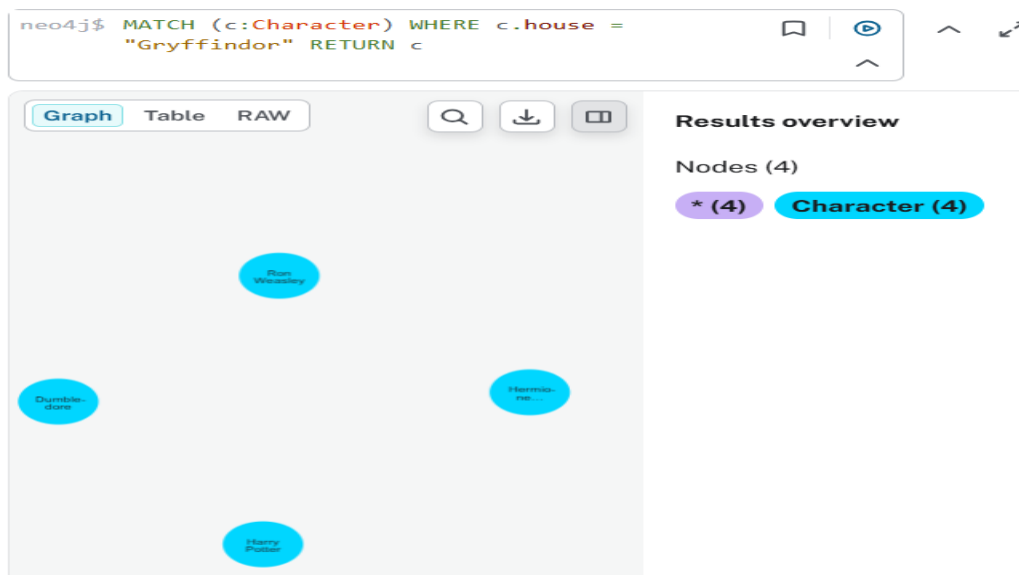
neo4j\$ MATCH (n)-[r]->(m) RETURN n, r, m

Graph Table RAW

	n	r	m
1	Harry Potter	FRIENDS_WITH	Hermione Granger
2	Harry Potter	FRIENDS_WITH	Ron Weasley
3	Ron Weasley	FRIENDS_WITH	Hermione Granger
4	Dumbledore	MENTORS	Harry Potter

MATCH (c:Character) WHERE c.house = "Gryffindor" RETURN c

The result:



```
neo4j$ MATCH (c:Character) WHERE c.house = "Gryffindor" RETURN c
```

Graph Table RAW

c

1	Harry Potter
2	Hermione Granger
3	Ron Weasley
4	Dumbledore

MATCH (a)-[r:MENTORS]->(b) RETURN a, r, b

The result:

```
neo4j$ MATCH (a)-[r:MENTORS]->(b) RETURN a, r, b
```

Graph Table RAW



Results overview

Nodes (2)

- * (2) Character (2)

Relationships (1)

- * (1) MENTORS (1)

neo4j> MATCH (a)-[r:MENTORS]-(b) RETURN a, b

GraphTableRAW

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	a	r	b
1	Dumbledore	MENTORS	Harry Potter